Orange3 project report

A supervised and unsupervised analysis of the Breast Cancer Wisconsin (Diagnostic) dataset with Orange3

Supervised and Unsupervised Learning for Biology (UE 5BM753)

Ann-Solène Adelson, Anaïs Hluszko and Corentin Osta

Year 2026/2027

Introduction

Breast cancer occurs when mammary epithelial cells acquire changes that disrupt normal proliferation and tissue organization. These changes also alter the appearance of the cell nucleus. Malignant cells typically are characterized by enlarged, irregular nuclei with abnormal chromatin. Pathologists use this nuclear atypia as an established sign when assessing a tumor[1,2].

When a breast mass is suspicious, a biopsy is needed to confirm the diagnosis. One common method is fine-needle aspiration (FNA), in which a thin needle collects cells for microscopic examination[1]. The sample is then interpreted by a pathologist. However, this interpretation is not perfectly reproducible. Indeed, in a large US study, pathologists agreed with the reference diagnosis in 75.3% of cases, and only in 48% of ambiguous, borderline cases[3]. A reproducible second opinion would therefore be useful. Measuring the size, shape and texture of the nuclei with image analysis, instead of judging them by eye, could provide one.

The Breast Cancer Wisconsin (Diagnostic) dataset (WDBC) puts this diagnostic setting in a quantitative form. It contains 569 samples, each one described from a digitized image of a fine-needle aspirate of a breast mass. For each sample, the contours and the grey-level appearance of the cell nuclei were converted into ten measurements: radius, texture, perimeter, area, smoothness, compactness, concavity, number of concave points, symmetry, and fractal dimension. Each measurement is summarized by its mean, its standard error, and a "worst" value, calculated from the most extreme measurements and therefore emphasizing the most atypical nuclei, which gives 30 continuous features[4,5]. Each sample is also labelled with its diagnosis, benign (B) or malignant (M).

The aim of this project is to determine whether malignant transformation leaves a nuclear signature strong enough to distinguish malignant from benign masses. We approach it in two complementary ways. In the supervised approach, the diagnosis is known for the examples used to train a model: we ask whether the 30 features can predict it, and which features have real discriminative power. In the unsupervised approach, the diagnosis is withheld while the cases are grouped: we ask whether the measurements alone form groups that correspond to the benign/malignant distinction.

To reach this aim, we structured this work in several sections. First, we present the dataset and the tools used for preprocessing and analysis (Orange3 and Python scripts), then the methodological steps: the comparison of classifiers and the selection of the most informative features, the clustering of the samples with four methods, and the test of the final model on a second cohort (WPBC). Next, we focus on the results: we show how well the diagnosis is predicted, which eight features are retained and what they measure biologically, how well the clusters match the diagnosis, and what the few errors on the external cohort reveal. Finally, in the Discussion, we compare the supervised and unsupervised approaches, interpret the selected features in light of the literature and of pathological grading, examine the influence of methodological choices on the results and state the limits of the project.

Materials and Methods

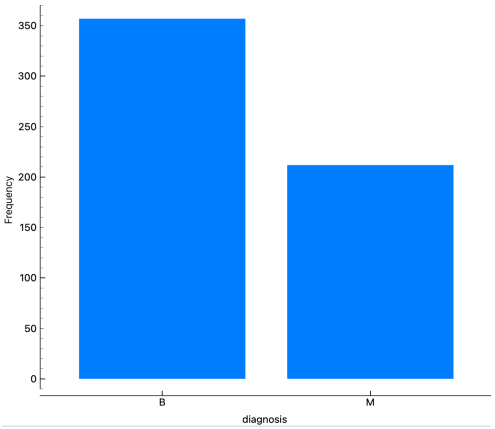
Dataset and preprocessing

Before asking models to distinguish diagnoses, we need to identify what each row and column represents and make the measurements comparable. The dataset was inspected using the *Data Info, Features statistics, Data table, Distribution, Correlations*, and *PCA* widgets of Orange3[6] (Figure A1). It contains 569 samples. The variable *diagnosis* is the target (B for

benign, M for malignant). The variable *id* only identifies the cases; it was only kept as metadata and was excluded from all preprocessing, feature selection, and modelling steps.

No values are missing in this dataset, so imputation did not change the data. An imputation step was nevertheless kept to make the workflow reusable on a dataset with missing values. It replaces a missing value by the average of the same feature (the default "Average/Most frequent option" of Orange3). No outliers were removed: extreme values (for example very large nuclei) are expected in malignant tumors and carry diagnostic information. The features were also standardized by centering them on their training mean and scaling them by their training standard deviation. This was necessary because they have different numerical scales. For example, the mean area is about 655, whereas the mean smoothness is about 0.1. Several algorithms, such as SVM, kNN, logistic regression and neural networks are sensitive to scale. In the supervised analysis, imputation and standardization were passed to *Test & Score* as preprocessing steps. They were therefore fitted separately within each cross-validation training fold, preventing information leakage from the validation fold[6].

Exploratory data analysis



Before selecting the modelling strategy, we first checked the class balance of diagnosis (*Distribution widget*, Figure 1): 357 tumors are benign (62.7%) and 212 are malignant (37.3%), or about 1.7 benign cases for every malignant case. This is a moderate imbalance, but it means accuracy alone could be misleading, because a model can appear accurate by favouring the more common class. We therefore used stratified cross-validation, and specific scores that focus on the malignant class to evaluate our models.

Figure 1. Class balance of the diagnosis variable. Bars show the number of samples in each class: 357 benign (B) and 212 malignant (M).

Then, we investigated the relationship among the predictor variables. We ranked the pairs of features by their Pearson correlation on the complete dataset (*Correlations widget*, Figure A2). Radius, perimeter and area are almost perfectly correlated, because they all describe the size of the nucleus. Some mean, worst and standard-error measurements of the same property are also strongly correlated. This motivated the feature selection described in the next section.

Supervised approach: classification models, feature selection and validation

Model	Reason for inclusion
Constant	Naïve reference model that always predicts the majority class
Logistic Regression	Simple and relatively interpretable linear baseline
SVM (RBF)	Effective classifier for continuous, potentially complex decision boundary
SVM (linear)	Same method with a straight boundary, to test whether the two classes are linearly separable
SVM (polynomial)	Same method with a curved boundary, as a simple nonlinear extension of the linear case
Random Forest	Ensemble model able to represent nonlinear effects and feature interactions
kNN	Distance-based method suitable for testing local separation between observations
Naive Bayes	Probabilistic baseline based on conditional independence assumptions
Classification Tree	Interpretable nonlinear decision model
Neural Network	Flexible nonlinear model, reported to reach the best accuracy (about 99%) on this dataset in a published comparison[7]

The first modelling question is whether the diagnosis (benign or malignant) can be predicted reliably from the 30 morphological features. A single algorithm cannot establish whether its assumptions suit the data, so we compared ten classifiers side by side (Table 1). They ranged from a trivial baseline to flexible nonlinear models.

This comparison provides both a performance floor and a way to assess whether more complex alternatives improve on simpler ones.

Table 1. Classifiers compared in the supervised approach. The table gives the reason for including each classifier.

To avoid overly optimistic results, a model must be evaluated on samples that were not used to fit it. The dataset was therefore divided once into a 70% training partition (399 samples) and a 30% final test partition (170 samples), using stratified sampling (*Data Sampler Widget," 11_Stratified_Split_70_30"*, Figure 2), so that the benign/malignant proportions were preserved in both subsets. All model screening, feature selection and hyperparameter tuning were performed using the training partition only. Candidate configurations were compared by stratified 10-fold cross-validation: the training set is divided into ten parts, each part is used once to evaluate a model trained on the nine others, and the results are averaged. The test partition was kept apart and used only once, after the final configuration was chosen.

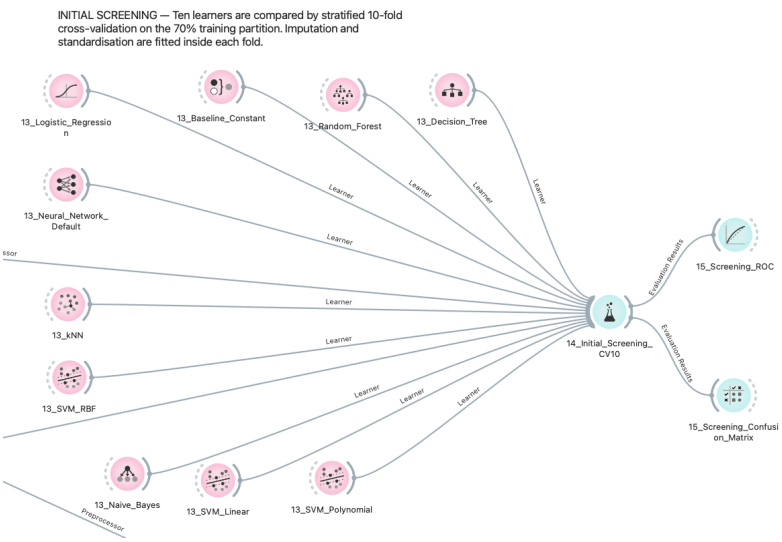


Figure 2. Supervised Orange3 workflow. The data are split 70/30. The ten classifiers are screened by stratified 10-fold cross-validation on the training part, features are selected, and the two final models are tested once on the held-out part. Preprocessing is fitted inside each training fold.

Prediction is not the only aim: we also want to know which of the 30 measurements are needed. Redundant features repeat the same information, and features with little predictive power add noise, so feature selection has two steps. First, when two features had a Pearson correlation above 0.95, only one of them was kept. This filter did not use the diagnosis; the correlations were computed on all 569 samples (Figure A2). Second, a logistic regression with an L1 penalty (C = 0.1) was fitted on the training set. The model gives a weight to each feature, and the L1 penalty shrinks the weights of unhelpful features to zero. Features with non-zero weights were kept, and the *Rank widget* was used to inspect these weights. As the filter and the L1 step did not use the same samples, we finally recomputed the correlations among the 8 retained features on the 399 training samples alone. The highest was 0.914, below 0.95, so no further features were removed (Figure A3).

The malignant class (M) was treated as the positive class. Models also have settings chosen before training, called hyperparameters. These were tuned using cross-validation on the training set only. For logistic regression, the regularization strength *C* was explored first with a broad search and then with a refined search, all on the training partition with 8 selected features. A smaller *C* penalizes complex models more strongly. For the neural network, six architectures and five values of the regularization parameter *alpha* were compared with the 30 features. The selected configuration was then refitted on the 8 selected features to provide a matched final comparison with logistic regression, but no additional hyperparameter search was performed for the neural network model on this reduced feature set. Models were assessed with seven scores (supervised scores in Table A1): AUC, accuracy, precision, recall,

specificity, F1, and MCC. Recall, the share of malignant tumors detected, received particular attention because missing a malignant tumor is clinically more serious than flagging a benign one for review. The final models were also described using their confusion matrix, which counts true and false positives and negatives. The parameters of the two final models are detailed in the appendix (Table A2).

Unsupervised approach: clustering methods, feature set and evaluation

The supervised analysis gives the model the diagnosis during training. A complementary question is whether the measurements form groups on their own. In clustering, samples with similar measurements are grouped without using the diagnosis, which is kept aside and used only afterwards to evaluate the clusters. All 569 samples were clustered (no train/test split), after imputation and standardization as in the supervised approach. We compared four clustering methods, because each one defines a group differently (Table 2).

Method	Reason for inclusion
K-means	Simple, fast and the standard reference method.
Hierarchical clustering (Ward)	Builds a tree that shows how the samples group at every level of similarity.
DBSCAN	Does not need the number of clusters in advance. Can spot outliers.
Spectral clustering	Can follow curved group shapes that K-means cannot.

Table 2. Clustering methods compared in the unsupervised approach. The table gives the reason for including each method.

K-means, hierarchical, and spectral clustering were asked to find two clusters, matching the two diagnoses, while DBSCAN determines the number of clusters itself. **K-means** places k centers in the data and assigns each case to the nearest center. We tried k from 2 to 7 and measured each result with the silhouette score, which shows how compact and well separated the clusters are. It was highest for $k = 2$, so k was fixed at 2. **Hierarchical clustering** starts with each sample alone and merges the closest groups step by step, which builds a tree. We used Ward linkage and the Euclidean distance, and cut the tree into two clusters. **DBSCAN** looks for dense regions of samples and labels a sample outside any dense region as noise. We required at least 5 neighbours and tested six neighbourhood distances (*eps*) from 2.0 to 2.6. **Spectral clustering** links each sample to its nearest neighbours and cuts this network into two groups. We tested 5 to 40 neighbours: the silhouette score hardly changed, so we kept 15 neighbours, which gave the best agreement with the diagnosis. This choice used the diagnosis and may make the spectral result slightly optimistic. We also tried a second way of linking samples, the RBF kernel, but it was not retained because it gave no conclusive results. Spectral clustering has no Orange widget, and the DBSCAN widget resets its distance each time it is opened, so both were run in Python Script widgets with scikit-learn[8].

The methods were run on two feature sets: first on all 30 features, then on the 8 features selected in the supervised approach (Figure 3). The second run used K-means, Ward and spectral clustering, with both the initial and the tuned settings for spectral clustering. The diagnosis was not given to the clustering in either run. It was kept only to evaluate the clusters afterwards. However, the 8 features were chosen with the help of the diagnosis. A better match with the diagnosis is therefore expected. The second run shows whether these features make the groups clearer. It does not show how well clustering works without any label. DBSCAN was not repeated on the 8 features because it found no useful structure with the 30 features.

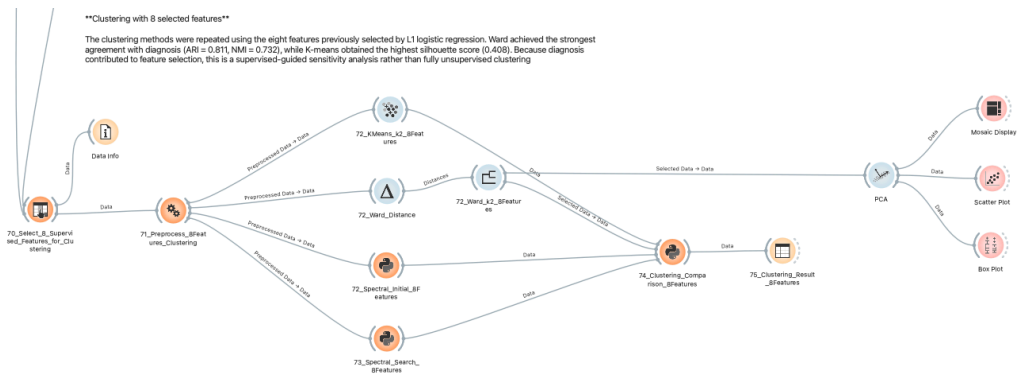


Figure 3. Orange3 workflow for the clustering with the 8 selected features. The 8 features chosen by the supervised approach are selected, imputed, and standardized, then clustered by K-means, Ward and spectral clustering. A python script widget compares the results.

Four scores were used (unsupervised scores, Table A1): the silhouette score, the adjusted Rand index (ARI), the normalized mutual information (NMI), and the best alignment. The silhouette score does not use the diagnosis; it only measures how compact and well separated the clusters are, so compact clusters may still not correspond to the diagnosis. ARI, NMI, and best alignment compare the clusters with the diagnosis. For DBSCAN, the number of cases labelled as noise was also counted. Mosaic displays and PCA plots coloured by cluster and by diagnosis provide visual comparisons.

Testing on a second cohort (WPBC)

To test the supervised model on other data, we used the Wisconsin Prognostic Breast Cancer (WPBC) dataset[9], which describes invasive breast cancers with the same nuclear measurements. Because it comes from the same research group as WDBC, some patients may appear in both datasets. Before the analysis in Orange, the WPBC dataset was pre-filtered using a Python script that removed the 162 samples whose patient identifier already appears in WDBC, which reduced the dataset from 198 to 36 samples. This filter relies on the identifier only, so samples recorded under a different identifier may remain, and the cohort is not guaranteed to be fully independent of the training data. It is needed because a patient present in both datasets would be evaluated on data the model has already seen, which would overestimate its performance, and would be counted twice in the clustering, which would distort the distances between samples.

These remaining 36 samples are described by the same 30 nuclear features (which required a mapping step in Python to align their slightly different names with the WDBC format), plus prognostic variables that we did not use. Because all 36 correspond to invasive cases, a 'diagnosis' feature fixed to 'M' was manually added to match the target structure of the original dataset. Only the recall can therefore be computed, because there are no benign samples to measure false positives, precision or specificity. The final logistic regression (8 features, $C = 0.5$) was applied to the 8 features of the WPBC samples with the *Apply Domain* and *Predictions widgets* (Figure A4). The model includes the standardization fitted on the training partition, so the WPBC samples were scaled with the training means and standard deviations, and nothing was refitted on WPBC. As a separate and exploratory step, the 36 WPBC samples were combined with the 569 WDBC samples (8 features, Concatenate widget), standardized together and grouped into two clusters by spectral clustering (15 neighbours). No score was computed, because WPBC has no benign samples. The step is only meant to look at where the WPBC samples fall relative to the benign and malignant groups.

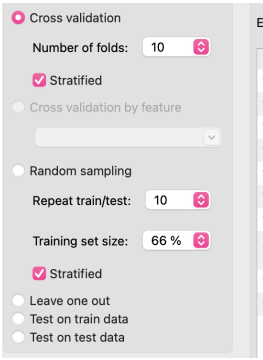
Results

Supervised approach

Logistic regression performs as well as more complex models.

A baseline model that always predicts “benign” reaches an accuracy of 0.627, but detects no malignant cases (recall of 0; Table 3). Accuracy alone is therefore not enough to judge a model on these data. Among the informative models, performance is very similar (Table 3). Logistic regression, the polynomial SVM, and the neural network detect the same proportion of malignant cases (recall of 0.966). Logistic regression achieves an AUC of 0.997 and an

accuracy of 0.980, values only 0.002 below those of the polynomial SVM. The decision tree is the weakest model, with a recall of 0.872. The ROC curves show the same overall pattern (Figure A5).



Evaluation results for target: M

Model	AUC	CA	F1	Prec	Recall	MCC	Spec
13_Logistic_Regression	0.997	0.980	0.973	0.980	0.966	0.957	0.988
13_Neural_Network_Default	0.996	0.972	0.963	0.960	0.966	0.941	0.976
13_SVM_polynomial	0.999	0.982	0.976	0.986	0.966	0.962	0.992
13_SVM_RBF	0.996	0.977	0.969	0.986	0.953	0.952	0.992
13_SVM_linéaire	0.997	0.965	0.953	0.953	0.953	0.925	0.972
13_Random_Forest	0.990	0.955	0.940	0.934	0.946	0.904	0.960
13_Naive_Bayes	0.986	0.945	0.927	0.915	0.940	0.883	0.948
13_kNN	0.988	0.960	0.944	0.978	0.913	0.914	0.988
13_Decision_Tree	0.899	0.920	0.890	0.909	0.872	0.828	0.948
13_Baseline_Constant	0.500	0.627	0.000	0.000	0.000	0.000	1.000

Table 3. Screening of the ten classifiers. Performance on the training set (399 samples), estimated by stratified 10-fold cross-validation. Precision, recall and F1 refer to the malignant class (M). Imputation and standardization were fitted within each fold.

With only 399 cases in the training set, these small differences do not establish a meaningful advantage for one model over another. When models perform similarly, the simpler model is generally preferable. Logistic regression is the simplest and most interpretable of these leading models: its coefficients show how each feature changes the odds of malignancy, and it requires neither a kernel nor hidden layers. The neural network, which the literature reports as the best model on this dataset[7], was kept as a second model, and both were compared on the test set.

Eight non-redundant features are enough for accurate diagnosis.

Feature selection reduces repeated information and shows which measurements contribute to the prediction. The correlation filter removed seven features that were almost copies of others (up to $r = 0.998$; Table A3) and left 23 features.

The L1-penalized logistic regression was then fitted on the training set. Eight of the 23 features retained non-zero weights. They converge on four dimensions of nuclear atypia. *Radius_mean* describes average nuclear enlargement, while *radius_se* reflects variability in nuclear size across the nuclei measured in a sample. *Concave_points_mean*, *concavity_worst*, *concave_points_worst*, and *smoothness_worst* quantify indentations and local irregularities of the nuclear contour. *Texture_worst* measures heterogeneity in grey-level intensity, an image-based proxy for variation in chromatin staining. Finally, *symmetry_worst* captures marked nuclear asymmetry. Together, these variables describe enlargement, variable shape (pleomorphism), membrane irregularity, and heterogeneous chromatin distribution, which are established cytological signs of malignancy[1,2].

This reduction had a small cost in cross-validation recall: with $C = 1$, recall was 0.966 using all 30 features (Table 3) and 0.946 using the 8 selected features. The final test-set evaluation below shows how the 8-feature model performed on held-out cases.

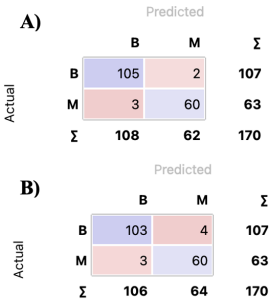
Logistic regression matches neural-network sensitivity with fewer false positives

The regularization strength C for logistic regression was tuned using the training set. Recall increased with C and then stabilised; we kept $C = 0.5$, which gave the best balance between recall and precision. A neural network was also tuned on the training set, but its regularization parameter had no measurable effect.

Model	AUC	CA	F1	Prec	Recall	MCC	Spec
35_LR_Final_C0.5_8Features	0.988	0.971	0.960	0.968	0.952	0.937	0.981
35_NN_Final_alpha0.1_8Features	0.987	0.959	0.945	0.938	0.952	0.912	0.963

Table 4. Scores of the two final models on the test set (170 samples). Both models use the 8 selected features. Precision, recall and F1 refer to the malignant class (M).

Both models were evaluated once on the same held-out test set of 170 observations (Table 4 and Figure 4). Logistic regression correctly classified 165 of 170 cases (97.1%). It made five errors: three false negatives, in which malignant tumors were classified as benign, and two false positives, in which benign tumors were classified as malignant. It detected 60 of the 63 malignant tumors (sensitivity of 95.2%) and correctly classified 105 of the 107 benign tumors (specificity of 98.1%).



Using the 8 selected features, the neural network detected the same 60 of 63 malignant tumors but produced 4 false positives, compared with 2 for logistic regression. Thus, on unseen test cases, it offered no improvement over logistic regression. Logistic regression remained the preferred model because it achieved the same sensitivity with fewer false positives and is simpler to interpret.

Figure 4. Confusion matrices obtained on the independent test set (170 samples). (A) Logistic regression with the 8 selected features (*C = 0.5*). (B) Neural network with the same 8 selected features. Rows indicate the actual diagnosis and columns the predicted diagnosis. A false negative is a malignant mass classified as benign; a false positive is the reverse.

Unsupervised approach

With the 30 features, spectral clustering matches the diagnosis best and DBSCAN does not.

With the 30 features, K-means and Ward each found two clusters. In these clusters, 90.5% and 88.0% of the samples matched their diagnosis (best alignment), with an ARI of 0.65 and 0.58 (Table 5A). Spectral clustering matched best: with 15 neighbours, 94.2% of the samples matched (ARI = 0.78), and the initial setting of 10 neighbours gave almost the same result. DBSCAN matched poorly. Its best setting (eps = 2.3) left 272 of the 569 samples (48%) as noise. So, only 52% of the samples were placed in the cluster with an ARI of 0.22. The other values of eps were no better: 202 to 371 samples stayed noise and the ARI remained between 0.10 and 0.22 (Table A4).

A)

	Method	Number of clusters	Noise instances	Coverage	Silhouette	ARI	NMI	Best alignment
1	Ward (k=2)	2	0	1	0.339385	0.575041	0.456915	0.880492
3	Spectral tuned (NN=15)	2	0	1	0.335552	0.779533	0.685242	0.942004
4	Spectral initial (NN=10)	2	0	1	0.336653	0.760801	0.662696	0.936731
2	K-means (k=2)	2	0	1	0.343382	0.653625	0.532408	0.905097
6	DBSCAN (eps=2.30, min_samples=5)	3	272	0.521968	0.317432	0.220014	0.211839	0.697715
5	DBSCAN (eps=2.00, min_samples=5)	4	371	0.347979	0.0977528	0.0963508	0.203526	0.637961

B)

	Method	Number of clusters	Noise instances	Silhouette	ARI	NMI	Best alignment
1	72_Ward_k2_8Features	2	0	0.397189	0.811164	0.732289	0.950791
2	72_KMeans_k2_8Features	2	0	0.408448	0.766827	0.682292	0.938489
3	72_Spectral_Initial_8Features	2	0	0.405053	0.792059	0.707124	0.945518
4	73_Spectral_Search_8Features	2	0	0.4029	0.792122	0.702238	0.945518

Table 5. Clustering results with (A) the 30 features and (B) the 8 selected features. Silhouette does not use the diagnosis. ARI, NMI and best alignment compare the clusters with the diagnosis (1 is perfect agreement). Noise instances are the samples that DBSCAN left outside any cluster; they count as one extra group in ARI, NMI and best alignment, and are excluded from the silhouette. Coverage is the share of samples placed in a cluster. In (A), DBSCAN is shown for eps = 2.3 (best ARI) and eps = 2.0, with at least 5 neighbours (other values in Table A4). DBSCAN was not run on the 8 features. NN: nearest neighbours.

The silhouette score is not enough to judge a clustering. It was about 0.34 for K-means, Ward, and spectral clustering, although their agreement with the diagnosis differs clearly (ARI from 0.58 to 0.78). DBSCAN with eps = 2.3 had a similar silhouette (0.317), because the silhouette ignores the noise samples, although almost half of the samples were left out and the ARI was only 0.22. This is why each clustering was also compared with the diagnosis.

Eight selected features sharpen the clusters, and Ward becomes the best method.

With the 8 features, all methods found two clusters and agreed better with the diagnosis (Table 5B). Ward reached the highest ARI (0.811), NMI (0.732) and best alignment (0.951). A best alignment of 0.951 means that about 541 of the 569 samples fall in the cluster that matches their diagnosis, and 28 do not. K-means had the highest silhouette (0.408), but a lower agreement with the diagnosis. The two spectral settings gave almost identical results (ARI = 0.792), so tuning the number of neighbours no longer mattered. The choice of features changed the partition more than the choice of settings.

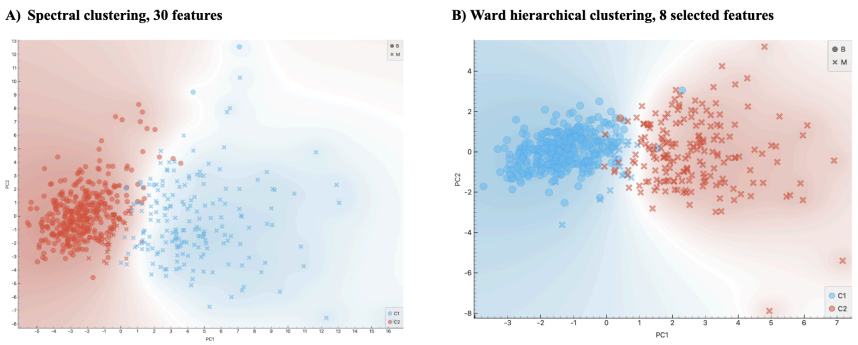


Figure 5. PCA projection of the 569 samples, coloured by cluster and shaped by diagnosis. The panels show the best method for each feature set, by best alignment with the diagnosis (Table 5). (A) Spectral clustering (15 neighbours) with the 30 features. (B) Ward clustering with the 8 selected features. Circles are benign (B) samples and crosses are malignant (M) samples. Cluster names are arbitrary: C1 is mostly malignant in A, and mostly benign in B. PC1 and PC2 are the first two principal components of the standardized features.

In the PCA plane (Figure 5), the two clusters are separated mainly along the first principal component, with a limited overlap at the boundary. A few malignant samples lie in the mostly benign cluster, and a few benign samples lie in the mostly malignant one. Mosaic displays of these clusters are shown in Figure A6.

External test on WPBC

The final model detects 34 of the 36 malignant WPBC samples.

The logistic regression (8 features, $C = 0.5$) detected 34 of the 36 WPBC samples, a recall of 94.4% (Figure 6), close to the sensitivity obtained on the test set (95.2%). It missed 2 samples (IDs 935058 and 935878). All WPBC samples are malignant, so only the recall can be read from this matrix. Both missed samples have low values of *concave_points_mean* and r*adius_se, below* the average of the combined data (Figure A7, panels A and B).

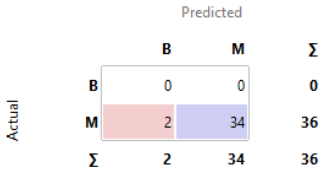


Figure 6. Confusion matrix of the logistic regression model on the 36 WPBC external cohort. All samples are malignant (M), so the benign (B) row is empty: 34 are correctly identified and 2 are missed (false negatives).

In an exploratory clustering, the two missed WPBC samples fall with the benign samples.

To see whether the 2 missed samples resemble benign samples, the 36 WPBC samples were combined with the 569 WDBC samples and grouped by spectral clustering (8 features, 15 neighbours, no label used). This method was chosen because it was the best with the 30 features and close to the best with the 8 features (Table 5). The two clusters separate mainly

along the first principal component (Figure 7). Four of the 36 WPBC samples fall in cluster C1, which contains mostly benign WDBC samples. They include the 2 samples missed by the logistic regression, while the other 2 were classified as malignant, so the two methods do not agree on every sample (Figure A7). The remaining 32 samples fall in C2, the mostly malignant cluster. A method that does not use the labels therefore also places the 2 missed samples next to benign samples, so these errors may not be random. The clusters were not scored, and the four samples lie close to the boundary between the clusters, where a small change of settings could move them.

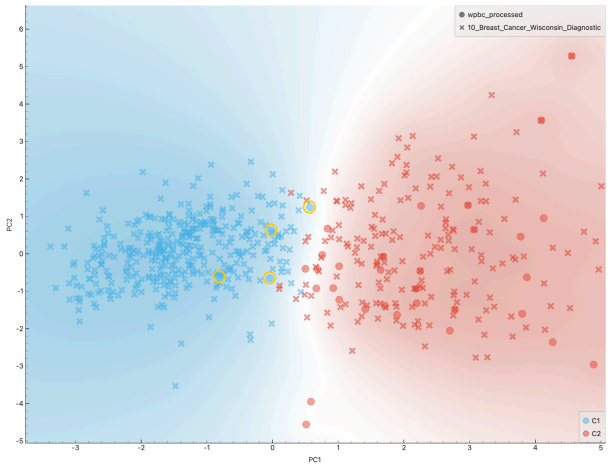


Figure 7. Exploratory spectral clustering of the full 569 WDBC samples and the 36 WPBC samples. PCA projection of the 8 selected features, standardized together. Circles are WPBC samples and crosses are WDBC samples. The colours are the two clusters found by spectral clustering (15 neighbours); their names are arbitrary. The 4 WPBC samples in the benign-like group (C1, blue) are circled in yellow. No score was computed. PC1 and PC2 are the first two principal components.

Discussion

The central question was whether nuclear measurements carry a signal strong enough to separate malignant from benign samples. Both approaches answer yes: a supervised model reached 97.1% accuracy on unseen samples, and clusterings built without any label matched the diagnosis for 94 to 95% of the samples. These two numbers are not strictly comparable, because they come from a test set and from the whole dataset respectively, but they point in the same direction.

Supervised approach

Eight nuclear measurements are enough to predict the diagnosis accurately. On 170 unseen cases, logistic regression detected 60 of the 63 malignant tumors and raised only 2 false alarms (accuracy 97.1%). This is close to the roughly 97% reported when this dataset was first analysed[4], although the evaluation methods differ. Several models performed almost equally well (Table 3), and the neural network brought no gain on the test set: it detected the same 60 tumors and raised more false alarms (4 against 2). We therefore retained logistic regression, the simplest and easiest model to interpret. A published comparison found the neural network to be the best model[7]. In our data, the best models are too close to be ranked and the difference may come from the data split or the tuning.

The selected features describe nuclear size, contour irregularity, texture and symmetry. These are the changes that pathologists use to recognize malignancy[1,2]. They also overlap with the three features of the first study (mean texture, worst area and worst smoothness)[4]. From a biological point of view, malignancy is not represented by nuclear enlargement alone: the model also relies on contour indentations, asymmetry and heterogeneous staining, so several aspects of nuclear organization change together. This agrees with pathological grading, where nuclear size, shape variability, chromatin appearance and membrane irregularity are assessed jointly[2]. The features are image measurements, not molecular markers, so they do not indicate which mutation or tumor subtype is involved.

Unsupervised approach and agreement between the approaches

Without any label, the measurements alone separate most benign from malignant samples. With 30 features, spectral clustering matched the diagnosis for 94% of the samples, whereas K-means and Ward, which look for compact groups, reached 88 to 91%. DBSCAN, which looks for dense regions separated by empty space, left 202 to 371 of the 569 samples as noise and agreed poorly with the diagnosis. This suggests that benign and malignant samples are not separated by a clear empty gap, which fits a gradual transition from benign to malignant nuclei rather than two sharply distinct groups. This conclusion is limited, because DBSCAN is sensitive to eps and we only tested six values. With the 8 selected features, all methods improved and Ward became the best (95%, ARI 0.81). This gain is partly expected, because the diagnosis guided the choice of these features, as it did for the 15 neighbours of spectral clustering.

External test and the two missed samples

On WPBC, the model detected 94.4% of the malignant samples, close to the 95.2% obtained on the WDBC test set. The model therefore seems to transfer to another cohort of invasive cancers, with the caveats given below. The 2 missed samples are informative. They have low values for the number of contour indentations and for the variability of nuclear size, and an unsupervised clustering that never saw the diagnosis also places them next to benign samples. Their errors are thus more likely to come from nuclei that look regular on these measurements than from a random failure of the model. A possible explanation, which we could not test, is that these two tumors are of low grade, so their nuclei have regular shapes and sizes, like those of benign samples. The two other WPBC samples in the benign-like cluster were classified as malignant, which shows that the model uses a combination of features and does not simply follow the clusters. WPBC also contains the tumor size and the lymph node status of each patient, which we did not use; checking whether the missed samples are small or low-grade tumors would be a natural next step.

Limits

Several limits should be kept in mind. The test set is small: detecting 60 of 63 tumors is compatible with a true detection rate between 87% and 98% (95% confidence interval). The WPBC cohort contains only 36 samples, all malignant, from the same research group as WDBC. Only the identifiers found in WDBC were removed, so the validation is not guaranteed to be fully independent of the training data, and with 36 samples the recall is imprecise (95% confidence interval of about 82 to 98%). WPBC has no benign samples, so specificity could not be tested, and the exploratory clustering of the combined samples (Figure 7) was not scored. The 8 features and the 15 neighbours were chosen with the help of the diagnosis, so the clustering results for these settings may be optimistic, and the two methods are not fully independent. Moreover, the dataset contains only nuclear features, whereas the diagnosis of a malignant tumor also relies on expressed receptors, genomic alterations, mitotic activity and the tumor microenvironment, which these measurements cannot capture.

In conclusion, a small set of nuclear measurements separates benign from malignant breast samples well, whether the diagnosis is used to train a model or not. Such a model could give pathologists a consistent second reading of an aspirate, but the errors we observed suggest it is least reliable for atypical or borderline cases. It should therefore remain a second opinion, and these ambiguous cases would require further biological analyses. A natural next step is to test the approach on a larger independent cohort that includes benign samples and difficult cases.

Contribution of each group member

The three members of the group designed the analytical strategy and the workflow together. **Ann-Solène Adelson** coordinated the group and was in charge of writing the report, preparing the figures and assembling the appendix with the material produced by the other members. She also checked the workflow, to make sure that it ran without errors and that the report matched it. **Anaïs Hluszko** led the technical side and built the workflow in Orange3. **Corentin Osta** built the external validation part on WPBC in Orange3. As they implemented the workflow, Anaïs and Corentin also proofread the report to confirm that the methods and results were described correctly from a technical point of view, and wrote some parts of it.

References

1. National Cancer Institute. *How Breast Cancer Is Diagnosed*. https://www.cancer.gov/types/breast/diagnosis
2. Bussolati, G., Marchiò, C., Gaetano, L., Lupo, R., & Sapino, A. (2008). Pleomorphism of the nuclear envelope in breast cancer: a new approach to an old problem. *Journal of Cellular and Molecular Medicine*, 12(1), 209–218. https://doi.org/10.1111/j.1582-4934.2007.00176.x
3. Elmore, J. G., Longton, G. M., Carney, P. A., Geller, B. M., Onega, T., Tosteson, A. N. A., Nelson, H. D., Pepe, M. S., Allison, K. H., Schnitt, S. J., O'Malley, F. P., & Weaver, D. L. (2015). Diagnostic concordance among pathologists interpreting breast biopsy specimens. *JAMA*, 313(11), 1122–1132. https://doi.org/10.1001/jama.2015.1405
4. Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. *Proceedings of SPIE*, 1905, 861–870.
5. UCI Machine Learning Repository. *Breast Cancer Wisconsin (Diagnostic)*. https://doi.org/10.24432/C5DW2B
6. Demšar, J., Curk, T., Erjavec, A., Gorup, Č., Hočevar, T., Milutinovič, M., Možina, M., Polajnar, M., Toplak, M., Starič, A., Štajdohar, M., Umek, L., Žagar, L., Žbontar, J., Žitnik, M., & Zupan, B. (2013). Orange: Data mining toolbox in Python. *Journal of Machine Learning Research*, 14, 2349–2353.
7. Agarap, A. F. M. (2018). On breast cancer detection: An application of machine learning algorithms on the Wisconsin diagnostic dataset. In *Proceedings of the 2nd International Conference on Machine Learning and Soft Computing (ICMLSC 2018)* (pp. 5–9). ACM. https://doi.org/10.1145/3184066.3184080
8. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
9. UCI Machine Learning Repository. *Breast Cancer Wisconsin (Prognostic)*. https://doi.org/10.24432/C5GK50

Appendices

The appendix gathers supporting material: the workflow screenshots, the correlation tables behind the feature selection, additional results and the definitions of the scores. Each item is cited in the main text.

A. Methods and scores

Score	Question it answers	Range
SUPERVISED APPROACH. Scores that compare the classifiers. Precision, recall and F1 refer to the malignant class (M).		
AUC	How well does the model rank malignant cases above benign ones, whatever the decision threshold?	0.5 (random) to 1 (perfect)
Accuracy (CA)	What share of all cases is classified correctly? It is misleading when the classes are unequal.	0 to 1
Precision	Among the cases predicted malignant, what share is truly malignant? A low value means many false alarms.	0 to 1
Recall (sensitivity)	Among the truly malignant cases, what share is detected? A low value means missed cancers.	0 to 1
Specificity	Among the truly benign cases, what share is correctly cleared?	0 to 1
F1 score	One score that balances precision and recall.	0 to 1
MCC	How well do predictions agree with the truth? It stays reliable when the classes are unequal.	−1 to 1 (0 = random)
UNSUPERVISED APPROACH. Scores that evaluate the clusters.		
Silhouette	How compact and well separated are the clusters? It does not use the diagnosis.	−1 to 1
ARI	How well do the clusters agree with the diagnosis, beyond chance?	0 (chance) to 1 (perfect)
NMI	How much information about the diagnosis do the clusters contain?	0 to 1
Best alignment	Share of cases whose cluster matches their diagnosis, after pairing each cluster with its best-fitting diagnosis. It is a description, not an accuracy.	0 to 1

Table A1. Scores used to compare the classifiers and to evaluate the clusters. The upper group (blue) compares the classifiers of the supervised approach; the lower group (green) evaluates the clusters of the unsupervised approach. Orange names precision and specificity "Prec" and "Spec" in its tables.

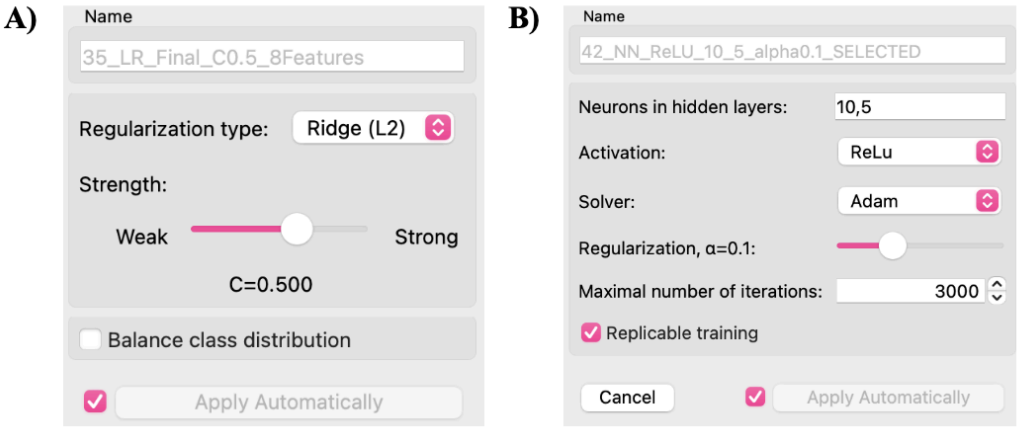


Table A2. Parameters of the two final models. A) Orange widget settings of the logistic regression (Ridge L2 penalty, C = 0.5, no class balancing). B) Orange widget settings of the neural network (two hidden layers of 10 and 5 neurons, ReLu activation, Adam solver, alpha = 0.1, 3 000 iterations, replicable training).

B. Data exploration and feature selection

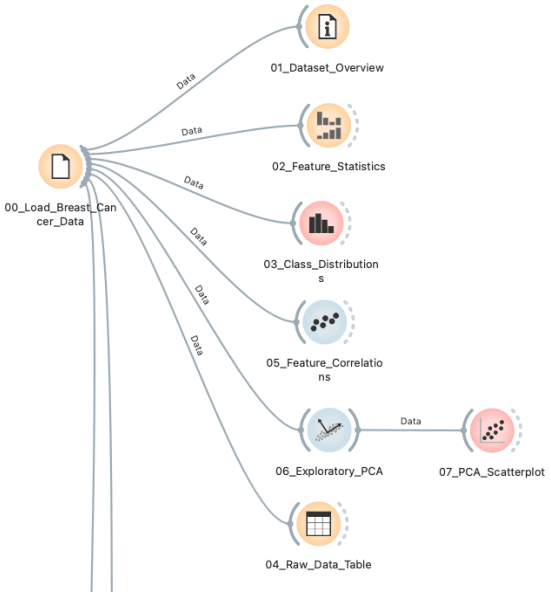


Figure A1. Orange3 workflow used to load and explore the dataset. The File widget loads the 569 samples and 30 features, with diagnosis as target and id as meta variable. The following widgets summarize the data and show its distributions, correlations and PCA projection. Arrows show the flow of data. The number at the start of each widget title gives its step in the workflow. Each widget is named by its type, with its title in the workflow in brackets: Data Info (01_Dataset_Overview), Column Statistics (02_Feature_Statistics), Distributions (03_Class_Distributions), Data Table (04_Raw_Data_Table), Correlations (05_Feature_Correlations), PCA (06_Exploratory_PCA), Scatter Plot (07_PCA_Scatterplot).

1	+0.998	perimeter_mean	:	radius_mean
2	+0.994	perimeter_worst	:	radius_worst
3	+0.987	area_mean	:	radius_mean
4	+0.987	area_mean	:	perimeter_mean
5	+0.984	area_worst	:	radius_worst
6	+0.978	area_worst	:	perimeter_worst
7	+0.973	perimeter_se	:	radius_se
8	+0.970	perimeter_mean	:	perimeter_worst
9	+0.970	radius_mean	:	radius_worst
10	+0.969	perimeter_mean	:	radius_worst
11	+0.965	perimeter_worst	:	radius_mean
12	+0.963	area_mean	:	radius_worst
13	+0.959	area_mean	:	area_worst
14	+0.959	area_mean	:	perimeter_worst
15	+0.952	area_se	:	radius_se
16	+0.942	area_worst	:	perimeter_mean
17	+0.941	area_worst	:	radius_mean
18	+0.938	area_se	:	perimeter_se

Figure A2. The 18 strongest correlations among the 30 features. Pearson correlation r on the 569 samples (Correlations widget). All pairs involve radius, perimeter or area, which all describe nuclear size; the first 15 are above 0.95.

1	+0.914	concave_points_mean	:	concave_points_worst
2	+0.855	concave_points_worst	:	concavity_worst
3	+0.823	concave_points_mean	:	radius_mean
4	+0.737	concave_points_mean	:	concavity_worst
5	+0.732	concave_points_worst	:	radius_mean
6	+0.697	concave_points_mean	:	radius_se
7	+0.693	radius_mean	:	radius_se
8	+0.541	concavity_worst	:	symmetry_worst
9	+0.536	concave_points_worst	:	smoothness_worst
10	+0.527	concave_points_worst	:	radius_se
11	+0.516	concavity_worst	:	smoothness_worst
12	+0.500	concavity_worst	:	radius_mean
13	+0.499	smoothness_worst	:	symmetry_worst
14	+0.480	concave_points_worst	:	symmetry_worst
15	+0.421	concave_points_mean	:	smoothness_worst
16	+0.421	concavity_worst	:	texture_worst
17	+0.409	concave_points_worst	:	texture_worst
18	+0.372	concavity_worst	:	radius_se

Figure A3. The 8 selected features are not redundant. Pearson correlations among the 8 selected features, on the 399 training samples. The 18 highest are shown; the maximum is 0.914, below the 0.95 threshold.

Redundant features

Radius, perimeter and area describe the same property, the size of the nucleus. Their values are almost perfectly correlated.

Group of features	Kept	Removed	Strongest correlation
Size of the nucleus (mean)	radius_mean	perimeter_mean, area_mean	r = 0.998
Variation in size (standard error)	radius_se	perimeter_se, area_se	r = 0.973
Size of the largest nuclei (worst)	none	radius_worst, perimeter_worst, area_worst	r = 0.994

Table A3. Groups of redundant features. The Pearson correlations were computed on the 569 cases, without using the diagnosis. The "worst" size features were removed because each is also strongly correlated with its "mean" version (r of 0.96 to 0.97).

C. External validation

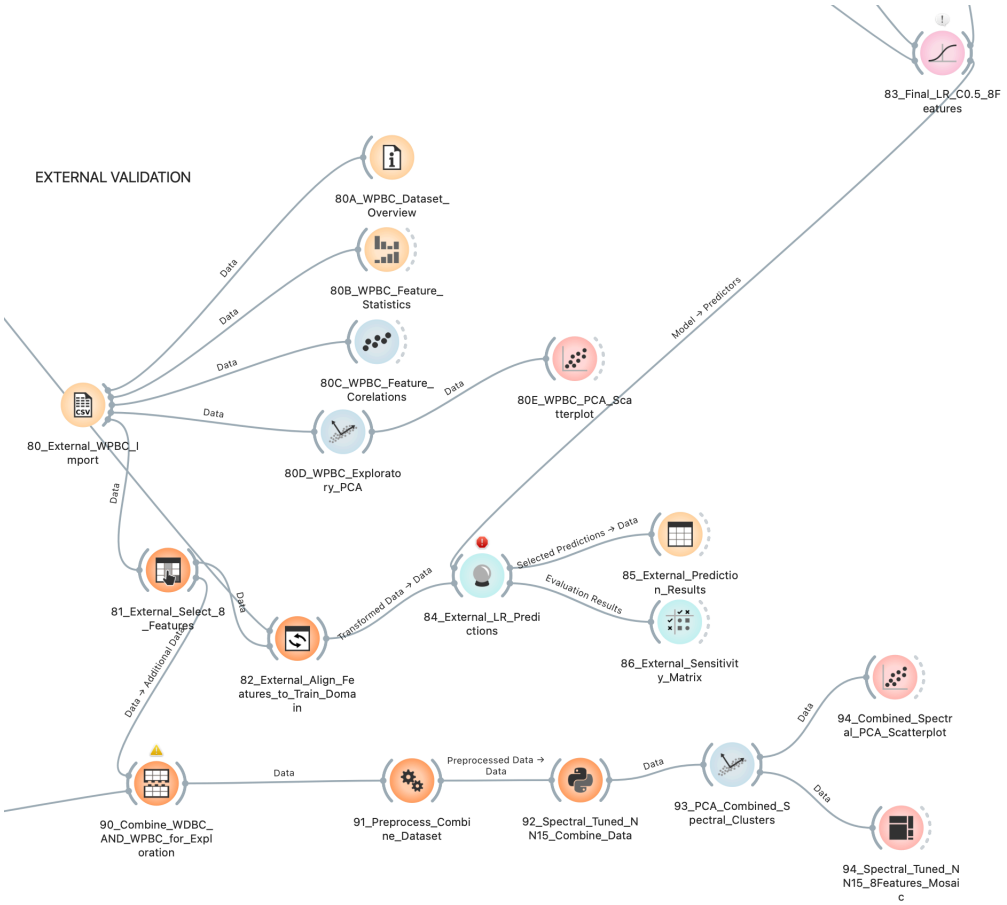


Figure A4. Orange3 workflow for the external validation on WPBC. The WPBC file is imported (80), its 8 features are selected (81) and put in the domain of the training data (82, Apply Domain), and the logistic regression model predicts their class (84 to 86: Predictions, Data Table, Confusion Matrix). The lower branch combines the WPBC samples with the 569 WDBC samples for the exploratory spectral clustering (90 to 94: Concatenate, Preprocess, Python Script, PCA, Scatter Plot). Widgets 80A to 80E describe the WPBC data.

D. Supervised screening

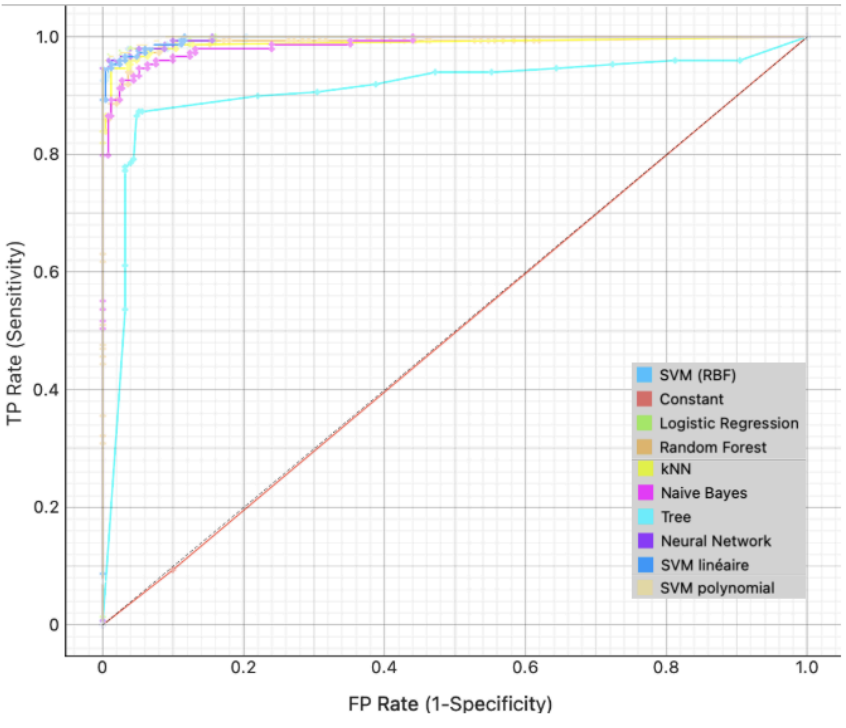


Figure A5. ROC curves of the ten classifiers on the training set. Stratified 10-fold cross-validation, 399 samples. The target class is M (malignant). The vertical axis is the share of malignant tumors detected. The horizontal axis is the share of benign tumors wrongly flagged as malignant. A curve close to the top left corner is better. The diagonal is the constant model. The lowest curve is the decision tree.

E. Clustering and exploratory clustering

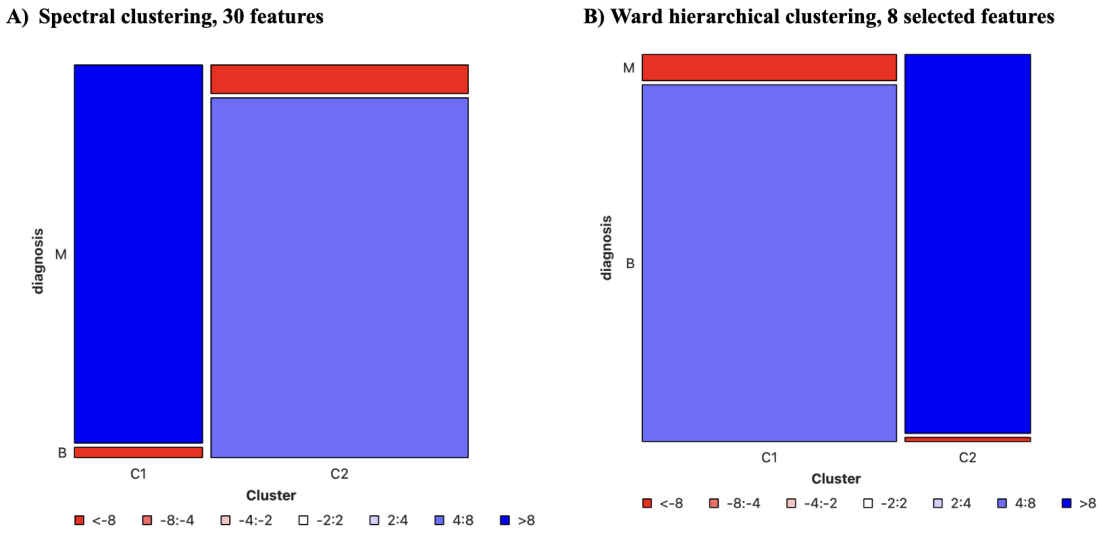


Figure A6. Mosaic displays of the clusters and the diagnosis. The two panels show the best clustering method for each feature set (Table 5). (A) Spectral clustering (15 neighbours) with the 30 features. (B) Ward clustering with the 8 selected features. How to read: each column is a cluster, and its width is the size of the cluster. Inside a column, the height of each tile is the share of benign (B) or malignant (M) samples in that cluster. The colour shows how far the tile is from what chance alone would give (standardized Pearson residual). Blue means more samples than expected, red means fewer. A good clustering shows large blue tiles and thin red tiles, as here. In B, the box is the Orange tooltip for the benign samples of C1 (355 observed against 239 expected).



Figure A7. The four WPBC samples placed in the benign-like cluster (C1) by the exploratory spectral clustering (Figure 7). Each panel is the Orange tooltip of one sample: its position on the first two principal components (PC1, PC2), its ID under Metas, and the standardized values (z-scores, computed on the combined WDBC and WPBC samples) of the 8 features used for clustering; 0 is the mean of the combined data. All four samples have a malignant diagnosis. Samples 935058 (A) and 935878 (B) are the two missed by the logistic regression; samples 927992 (C) and 938413 (D) were classified as malignant.

F. DBSCAN with different neighbourhood distances

	Configuration	Epsilon	Min samples	Number of cluster	Noise instances	Coverage	ARI	NMI	Best alignment
4	DBSCAN-e2.30-m5	2.3	5	3	272	0.521968	0.220014	0.211839	0.697715
5	DBSCAN-e2.40-m5	2.4	5	1	248	0.564148	0.20186	0.145187	0.725835
3	DBSCAN-e2.20-m5	2.2	5	4	312	0.45167	0.175545	0.228998	0.657293
6	DBSCAN-e2.60-m5	2.6	5	1	202	0.644991	0.173636	0.108519	0.711775
2	DBSCAN-e2.10-m5	2.1	5	3	347	0.390158	0.149152	0.232291	0.667838
1	DBSCAN-e2.00-m5	2	5	4	371	0.347979	0.0963508	0.203526	0.637961

Table A4. DBSCAN with different neighbourhood distances (eps). Minimum 5 neighbours, 30 features standardized, Euclidean distance. Noise samples count as one extra group in ARI, NMI and best alignment. Coverage is the share of samples placed in a cluster. The DBSCAN widget of Orange3 resets its distance each time it is opened, so each value was run in a Python Script widget (scikit-learn) with a fixed eps.